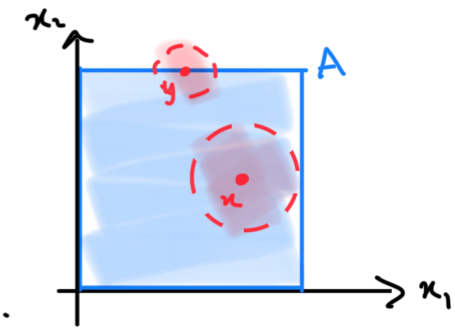
Def Let (X, d) be a metric space and $A \subseteq X$. We say:

① $a \in X$ is an <u>interior point</u> of A if $\exists\, r > 0$ s.t. $B_r(a) \subseteq A$.

② $A^\circ = \{a \in X : a \text{ is an interior point of } A\}$ is the <u>interior of A</u>.

③ A is <u>open</u> if $A^\circ = A$.

Ex $X = \mathbb{R}^2$ with the metric d_2, and $A = [0,1] \times [0,1]$.

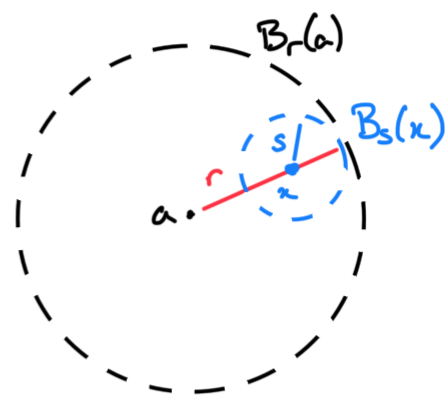


- $x = (\frac{3}{4}, \frac{1}{2})$ is an interior point of A: $B_{1/4}(x) \subseteq A$
- $y = (\frac{1}{2}, 1)$ is not an interior point of A: $\forall r > 0$, $B_r(y) \not\subseteq A$.
- $A^\circ = (0,1) \times (0,1)$
- A is not open in $(\mathbb{R}^2, d_2)$.

Ex Let (X, d) be a metric space.

- $\emptyset$ and X are open sets.
- $B_r(a)$ is an open set $\forall a \in X$ and $\forall r > 0$:

Claim: $B_r(a) \subseteq (B_r(a))^\circ$.



Fix $x \in B_r(a)$, i.e. $d(x,a) < r$.

Want: $x \in (B_r(a))^\circ$, i.e. $\exists\, s > 0$ s.t. $B_s(x) \subseteq B_r(a)$. Set

$s = r - d(x,a) > 0$. Then

$$y \in B_s(x) \implies d(y,x) < s$$
$$\implies d(y,a) \le d(y,x) + d(x,a)$$
$$< s + d(x,a) = r$$

So $B_s(x) \subseteq B_r(a)$.

<u>Rmk</u> Note that $A^\circ \subseteq A$ for any set A:

$$x \in A^\circ \implies \exists\, r > 0 \text{ s.t. } B_r(x) \subseteq A$$
$$\implies x \in B_r(x) \subseteq A$$

since $d(x,x) = 0 < r$

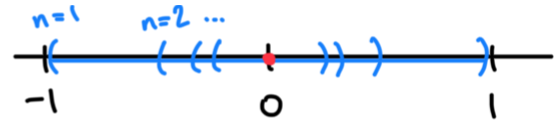
So, to prove that a set is open, it suffices to show $A^\circ \supseteq A$.

<u>Prop</u> Let (X,d) be a metric space and $A, B \subseteq X$.

1. If $A \subseteq B$ then $A^\circ \subseteq B^\circ$.
2. $A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$
3. $A^\circ \cap B^\circ = (A \cap B)^\circ$
4. $(A^\circ)^\circ = A^\circ$. In particular, A° is an open set.
5. A° is the largest open set contained in A.
6. A finite intersection of open sets is open.
7. An arbitrary union of open sets is open.

<u>Rmk</u> An arbitrary intersection of open sets need not be open. Eg., in $(\mathbb{R}, |\cdot|)$,

$$\bigcap_{n\in\mathbb{N}} \underbrace{\left(-\tfrac{1}{n}, \tfrac{1}{n}\right)}_{= B_{1/n}(0)} = \underbrace{\{0\}}_{\text{not open:}}$$



open $\forall n$

$\forall r>0, \quad B_r(0) = (-r, r) \not\subseteq \{0\}$

Pf: ① Suppose $A \subseteq B$. Want: $A^\circ \subseteq B^\circ$.

$$a \in A^\circ \Rightarrow \exists r>0 \text{ s.t. } B_r(a) \subseteq A$$
$$\Rightarrow B_r(a) \subseteq A \subseteq B \Rightarrow a \in B^\circ.$$

②
$$\left.\begin{array}{l} A \subseteq A\cup B \overset{①}{\Rightarrow} A^\circ \subseteq (A\cup B)^\circ \\ B \subseteq A\cup B \overset{①}{\Rightarrow} B^\circ \subseteq (A\cup B)^\circ \end{array}\right\} \Rightarrow A^\circ \cup B^\circ \subseteq (A\cup B)^\circ$$

③
$$\left.\begin{array}{l} A\cap B \subseteq A \overset{①}{\Rightarrow} (A\cap B)^\circ \subseteq A^\circ \\ A\cap B \subseteq B \overset{①}{\Rightarrow} (A\cap B)^\circ \subseteq B^\circ \end{array}\right\} \Rightarrow (A\cap B)^\circ \subseteq A^\circ \cap B^\circ$$

Want: $A^\circ \cap B^\circ \subseteq (A\cap B)^\circ$. Fix $x \in A^\circ \cap B^\circ$.

$$x \in A^\circ \Rightarrow \exists r_1 > 0 \text{ s.t. } B_{r_1}(x) \subseteq A$$
$$x \in B^\circ \Rightarrow \exists r_2 > 0 \text{ s.t. } B_{r_2}(x) \subseteq B$$

Set $r = \min\{r_1, r_2\} > 0$. Then

$$y \in B_r(x) \Rightarrow d(y,x) < r$$
$$\Rightarrow d(y,x) < r_1 \text{ and } d(y,x) < r_2$$
$$\Rightarrow y \in B_{r_1}(x) \text{ and } y \in B_{r_2}(x)$$

So $B_r(x) \subseteq B_{r_1}(x) \cap B_{r_2}(x) \subseteq A\cap B$, and thus $x \in (A\cap B)^\circ$.

④ Want: $(A^\circ)^\circ = A^\circ$. Recall that $A^\circ \subseteq A$ for any

set A. So $(A^\circ)^\circ \subseteq A^\circ$ by ①.

Claim: $A^\circ \subseteq (A^\circ)^\circ$.

$$x \in A^\circ \Rightarrow \exists r > 0 \text{ s.t. } B_r(x) \subseteq A$$
$$\overset{①}{\Rightarrow} \underset{\substack{\| \\ B_r(x)}}{(B_r(x))^\circ} \subseteq A^\circ, \quad \text{since } B_r(x) \text{ is open}$$
$$\Rightarrow x \in (A^\circ)^\circ.$$

⑤ Want: B open, $B \subseteq A \Rightarrow B \subseteq A^\circ$.

- A° is the largest open subset of A: any other $B \subseteq A$ that is open must be contained in A°.

$$B \subseteq A \overset{①}{\Rightarrow} \underset{\substack{\| \\ B}}{B^\circ} \subseteq A^\circ, \quad \text{since } B \text{ is open}$$

⑥ Claim: A, B open $\Rightarrow A \cap B$ open.

$$(A \cap B)^\circ \underset{③}{=} A^\circ \cap B^\circ \underset{A,B \text{ open}}{=} A \cap B$$

So:

$$\left.\begin{array}{l} A_1, A_2 \text{ open} \Rightarrow A_1 \cap A_2 \text{ open} \\ \qquad\qquad\qquad\quad A_3 \text{ open} \end{array}\right\} \Rightarrow \left.\begin{array}{l} (A_1 \cap A_2) \cap A_3 \text{ open} \\ \qquad\qquad\quad A_4 \text{ open} \end{array}\right\} \cdots$$

A simple induction argument proves: $\forall n \in \mathbb{N}$,

$A_1, \ldots, A_n$ open $\Rightarrow A_1 \cap \ldots \cap A_n$ open.

- "Finite intersection" means an intersection of finitely many sets

⑦ Let I be a set. Suppose A_i open $\forall i \in I$.

Want: $\bigcup_{i \in I} A_i = \{x \in X : \exists i \in I \text{ s.t. } x \in A_i\}$ is open.

- "Arbitrary union" means a union of finitely or infinitely many sets

Fix $x \in \bigcup_{i \in I} A_i$. Then $\exists i_0 \in I$ s.t. $x \in A_{i_0}$. As A_{i_0} is open, $\exists r > 0$ s.t. $B_r(x) \subseteq A_{i_0}$. Then $B_r(x) \subseteq \bigcup_{i \in I} A_i$, and so $x \in \left(\bigcup_{i \in I} A_i\right)^\circ$. So $\bigcup_{i \in I} A_i$ is open. $\square$